

Thomas Arocena

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Education

M2	Université de Lorraine , M2 in Fundamental and Applied Mathematics – Nancy, France	Sept 2026 – Aug 2027
	• Continuous-time and discrete-time stochastic process • Graph theory • Stochastic control	
	Ecole des mines de Nancy , Mathematical Engineering (Engineering Degree) – Nancy, France	Sept 2023 – Aug 2027
	• Monte Carlo method • Partial differential equation : analytical and numerical solutions • Algorithms and computer science	
	Lycée Kléber , CPGE MPSI/MP* – Strasbourg, France	Sept 2021 – Aug 2023

Experience

Gustave Roussy , Statistics Intern – Villejuif, France	Jan 2026 – Mar 2026
Comparison of sample size calculation software for different clinical trial designs	
• Developed an R pipeline to extract and compare data from proprietary software • Visualized, analyzed, and interpreted sample size calculations based on statistical formulas and clinical considerations • Developed an interactive Quarto website to present and share the results	
Spotlight Medical , Research Intern – Paris, France	June 2025 – Dec 2025
• Conducted survival analysis on real-world clinical data and developed models for visualization and prediction • Used R Shiny and golem to develop an R package and interactive applications for clinical data visualization, analysis, and modeling	

Projects

Oncostat's SSC	Jan 2026 – Mar 2026
Open-source website for comparing sample size calculation methods and software	
eBPF in Lean 4	2024
Formalization of eBPF in Lean 4 (student project)	
• Formalized parts of the eBPF instruction set in Lean 4 • Explored formal verification and theorem proving through a student research project	
TIPE 2023	2023
Modeling urban resilience to flooding as part of the TIPE (student project)	

Skills

Mathematics: Probability, Statistics, Algebra, Calculus

Languages: French, English (C1)

Programming Languages: Python, R, MATLAB, VBA, LaTeX, Markdown, Lean 4